

Concept of the Section “Common Infrastructures” - section-infra

National Research Data Infrastructure (NFDI) e.V.

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Abstract

The section organizes the coordination of the NFDI consortia and other actors for the development of federated infrastructures and software components and is thus also the central point of contact for the development of basic services. In addition to the identification, conception and development of jointly usable infrastructure components and their interoperability, the section has the specific goal of developing a Research Data Commons (RDC) - a commons for research data. Another goal is to establish sustainable structures for technology partnerships within the NFDI in order to organize the provision of shared information infrastructures in the long term. In recent years, the section has created a functioning internal organization (despite a lack of funding). The working groups that have been set up (currently 11) have largely proved to be very successful; the majority of them are currently working on the preparation and development of basic services. However, the lack of an architectural concept supported by all consortia proved to be a serious deficit. This was addressed in the section with the creation of a corresponding working group. This will improve the possibility of recognizing deficits and overlaps in the working groups in the future and thus achieve a better focus and division of labour. In addition to providing support for RDC development, the cross-sectional tasks of the section include, above all, those that improve the organizational structure and communication with the working groups, other sections and consortia. The overview and planning of further activities will in future be supported by internal documentation and a knowledge base as well as an outreach strategy to be developed. The section is involved in various initiatives outside the NFDI. Many stakeholders are not only involved in the NFDI, but are also active in other (inter)national networks, standardization initiatives and projects. There are synergies with the ESFRI Landmarks and projects, the FAIR Data Spaces project (GAIA-X), the Research Data Alliance (RDA) and the European Open Science Cloud (EOSC). In particular, the section will support the establishment of the NFDI as a national EOSC node.

Introduction

The section organizes the coordination of the NFDI consortia and other players for the **development of federated infrastructures and software components**. It is therefore the central point of contact for the development of basic services, i.e. components of a basic infrastructure for potentially all consortia. Offers of this kind that technology partners would like to develop for the NFDI should be synchronized with the structured work process outlined here as early and continuously as possible. Coordination with the other sections is also important here.

The section concept presented here specifies areas of work for the development of jointly used and networked information infrastructures and names concrete elements that are required for the desired interoperability within the NFDI and beyond. The NFDI consortia bring with them a large number of heterogeneous and distributed research data infrastructures that are only networked to a limited extent. A central requirement for the section is to identify components for overarching use and to develop a proposal on how these can be set up and implemented in technical terms.

Specifically, the section will work on the realization of a federated basic infrastructure. The so-called “**Research Data Commons**” (RDC) should enable uniform access to data, software and compute resources as well as sovereign data exchange and collaborative work¹. The RDC concept has become increasingly important internationally for the development of research data infrastructures: Examples include the Australian Research Data Commons (ARDC)², the US National Cancer Institute Research Data Commons (NCI RDC)³ or the European Open Science Cloud (EOSC). Some NFDI consortia also name an RDC as an explicit goal and support the necessary work with their resources. The RDC introduces a paradigm shift from data to function shipping.

Furthermore, the section aims to lay the foundations for **technical and organizational structures** in which the sustainable provision of shared information infrastructures can take place in the long term. Collaboration is practiced in the section and a functioning internal organization is created. The technical content is developed by working groups in which representatives of the consortium initiatives and other technical partners contribute know-how, personnel, software and computing resources. A participatory culture and iterative working methods should help to ensure that stakeholders from all NFDI consortia are well integrated and that parallel developments outside the NFDI are appropriately addressed.

Status quo and retrospective

The desire and need to establish federated infrastructures and services was already formulated by a large number of the consortia in the “Berlin Declaration”⁴ and the “Leipzig-Berlin Declaration”⁵ before funding began. The cross-cutting topics mentioned in the papers were prioritized, bundled into initial sections and officially confirmed by the Scientific Senate in October 2021. The “Common Infrastructures” section was also founded. Subsequently, topic-specific working groups were set up in the section. Some of these working groups proved to be very successful. For example, the “Identity and Access Management” working group and the “Persistent Identifiers” working group were able to draw on many years of preliminary work in the national and international environment. Other WGs such as “Multicloud” or “Data Integration” proved to be much more complex.

In addition, the commitment of consortium personnel to the WGs proved to be difficult due to the funding cuts in the first phase of the consortia. The establishment of another consortium - Base4NFDI, which exclusively supports the development of basic services, was helpful in this

¹ Grossman, R.L. Ten lessons for data sharing with a data commons. *Sci Data* 10, 120 (2023). <https://doi.org/10.1038/s41597-023-02029-x>

² Barker, M., Wilkinson, R. & Treloar, A. The Australian research data commons. *Data science journal* 18 (2019). <https://doi.org/10.5334/dsj-2019-044>

³ NCDI Cancer Research Data Commons. <https://datascience.cancer.gov/data-commons> (2023-04-24)

⁴ Glöckner, Frank Oliver, Diepenbroek, Michael, Felden, Janine, Overmann, Jörg, Bonn, Aletta, Gemeinholzer, Birgit, Güntsch, Anton, König-Ries, Birgitta, Seeger, Bernhard, Pollex-Krüger, Annette, Fluck, Julianne, Pigeot, Iris, Kirsten Toralf, Mühlhaus, Timo, Wolf, Christof, Heinrich, Uwe, Steinbeck, Christoph, Koepler, Oliver, Stegle, Oliver, Weimann, Joachim; Schörner-Sadenius, Thomas; Gutt, Christian; Stahl, Florian; Wagemann, Kurt; Schrade, Torsten; Schmitt, Robert; Eberl, Chris; Gauterin, Frank; Schultz, Martin; Bernard, Lars. (2019). Berlin Declaration on NFDI Cross-Cutting Topics (Version 1). Zenodo. <https://doi.org/10.5281/zenodo.3457213>

⁵ Ebert, B., Fluck, J., Glöckner, F. O., Koepler, O., Miller, B., Schmitt, R., Schrade, T., Stegle, O., Steinbeck, C., von Suchodoletz, D., Wagemann, K., Knebes, J., Kraft, S., Seitz-Moskaliuk, H., Sure-Vetter, Y., & Wössner, E. (2021). NFDI Cross-cutting Topics Workshop Report. Zenodo. <https://doi.org/10.5281/zenodo.7825374>

respect. The “Common Infrastructures” section and its working groups were intensively involved in the application process and contribute significantly to the work in Base4NFDI.

The lack of an **architectural concept** supported by all consortia proved to be a serious shortcoming. Although the Research Data Commons (RDC) has already established an initial conceptual framework based on the international model, the technical and organizational implementation is not very concrete and has not been coordinated between the consortia. Following initial discussions between the sections and the Directorate in 2023, the **“Overall Architecture” working group** was therefore established at the beginning of 2024. This improves the possibility of recognizing deficits and overlaps in the WGs and thus achieving a better alignment and division of labor.

In particular, a consensus architecture concept also enables better localization and classification of basic services that are currently being developed via Base4NFDI. A subdivision into core components and peripheral services - which tend to be located at the level of data production, processing or evaluation - makes it easier to decide for which components particular sustainability is required. The issue of sustainability is closely linked to the governance of a federated data infrastructure. This is addressed in all WGs and basic services alongside technology and organization, especially in the “Overall Architecture” WG. The work of the central NFDI **Task Force “Governance and Sustainability”** founded in 2024 is seen as a necessary complementary contribution.

Goals and success criteria

The aim of the NFDI is to create a framework for an interdisciplinary, shared research data space in which information is collected, organized and made available in accordance with the FAIR principles⁶. Accordingly, the section pursues three goals:

1. **Networking of infrastructure components for shared use:** the identification, conception and division of labor in the development of jointly usable infrastructure components and their interoperability.
2. **Research Data Commons - a commons for research data:** The creation of a multi-cloud-based infrastructure that can be used by consortia to exchange and share data, software, and compute resources.
3. **Sustainable structures:** The establishment of sustainable structures for technology partnerships within the NFDI in order to organize the provision of shared information infrastructures in the long term.

Networking of infrastructure components for shared use (Z1)

The NFDI consortia bring with them a large number of heterogeneous and distributed information infrastructures and resources that are only networked with each other to a limited extent. A central requirement for the section is to identify components for joint use and to draw up a proposal on how these can be set up and implemented in technical terms and which requirements are necessary for networking. The entire data life cycle is affected.

⁶ <https://www.nfdi.de/wp-content/uploads/2023/03/Einseiter-Konsortien.pdf>

Networking requires interoperability at the technical, semantic, organizational and legal levels. Agreement on protocols and standards and the specification of corresponding workflows and interfaces are just as necessary as compliance with legal standards and security standards. The “EOSC Interoperability Framework - EIF”⁷ serves as an important conceptual template.

According to our current understanding, the concrete infrastructure components comprise a series of elements that can be declared as so-called “basic services” - in line with the NFDI Expert Commission, we understand this to mean services that guarantee the basic supply of infrastructure for potentially all consortia or in the use of which there is a common interest within the consortia and for which competing or incompatible solutions should be avoided.⁸ Harmonization among users and providers is essential for the development of such basic services. In doing so, the section is building on one of the strengths of the NFDI: its members have decades of experience in providing centralized services. This experience is often well-tested in specific domains, but varies greatly between domains. In addition, the section has received substantial support from the Base4NFDI network since 2023. Some basic services (e.g. IAM4NFDI, PID4NFDI) have already been launched from the section working groups and others are in preparation. The development of basic services is subject to an orderly process involving all NFDI committees.

Not all WGs necessarily result in basic services. Here it is important to maintain at least one forum via the section and the corresponding working groups, which leads the needs and approaches of the consortia to solutions that are as convergent as possible. Such topics in particular also require cooperation with other sections. In order not to compromise functioning domain infrastructures, the connection to joint structures should be made possible and not forced (“opt-in” model).

Finally, the topic of provenance, i.e. the traceability of the origin and changes to scientific data and software, is of particular importance. In close cooperation with the “Metadata, Terminologies and Provenance” section, the section will work on requirements for the IT-supported documentation of data in research processes, from generation and processing to final use or re-use, and will deal with the long-term availability and accessibility of data. There are still different requirements in the scientific domains.

Research Data Commons (Z2)

The section follows the principle that is also the guiding principle for the NFDI concept as a whole: existing resources should be reused and intelligently connected wherever possible. The specific goal - as a “proof of principle”, so to speak - is to implement a concrete federated basic infrastructure that makes this possible.

⁷ European Commission: Directorate-General for Research and Innovation, Corcho, O., Eriksson, M., Kurowski, K., Ojsteršek, M. et al., *EOSC interoperability framework – Report from the EOSC Executive Board Working Groups FAIR and Architecture*, Publications Office, 2021, <https://data.europa.eu/doi/10.2777/620649>

⁸ Vgl. NFDI-Ausschreibung 2021,

https://www.dfg.de/foerderung/info_wissenschaft/info_wissenschaft_21_37/index.html sowie

DFG - Deutsche Forschungsgemeinschaft (2020): Der Aufbau einer Nationalen Forschungsdateninfrastruktur. Zweite Stellungnahme des NFDI-Expertengremiums.

https://www.dfg.de/download/pdf/foerderung/programme/nfdi/stellungnahme_nfdi_201112_de.pdf.

These so-called “Research Data Commons” (RDC) were supported by the majority of prospective consortia in 2019/2020.⁹ As part of the section's work, a multi-cloud-based infrastructure is to be created that can be used by the consortia to exchange and share data, software and compute resources. Hybrid solutions and edge components are not excluded. In addition to shared cloud services as the base layer, the NFDI-RDC concept essentially comprises collaborative workspaces for the integration and reuse of data, HPC access and a shared authentication and authorization infrastructure (AAI) (Fig. 1).

The NFDI-RDC thus forms a bridge between service and data providers and consumers. It will have a significant effect on the quality and openness of data and is expected to significantly increase efficiency in the mobilization and use of existing data. Overall, this provides strong incentives for potential users of NFDI offerings.

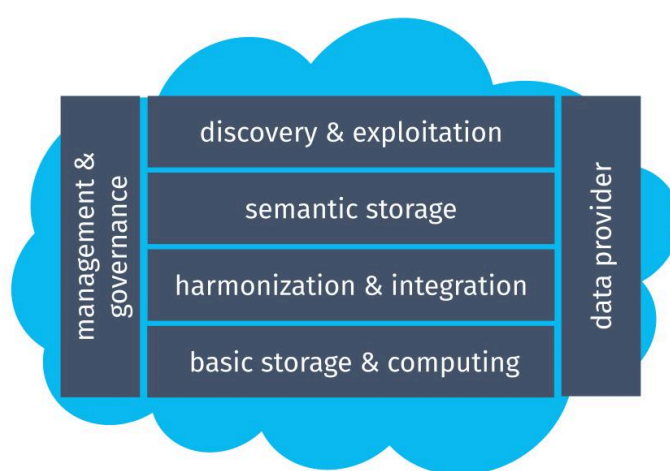


Figure 1: Research Data Commons - simplified draft of an RDC architecture.

The Research Data Commons (RDC) represents a conceptual framework for a federated NFDI basic infrastructure. With the establishment of the “Overall Architecture” working group, the possibility was created to concretize the technical and organizational implementation and to coordinate it between the consortia. A consensus architecture concept provides the “guard rails” for both the WGs and the resulting basic services.

Among the architecture concepts currently under discussion, the data mesh concept¹⁰ is particularly noteworthy. Considering the heterogeneous, complex and dynamic requirements of the consortia, Data Mesh has considerable potential for the entire NFDI. The approach of decentralization and domain-specific autonomy of data offers an alternative to the constraints of traditional architectures. However, a successful development partnership within the NFDI is critical to achieving the objectives.

The development of the RDC is designed as a long-term iterative process. In order to avoid undesirable developments, it makes sense to take well-defined and manageable steps that are reviewed and approved by the NFDI committees.

⁹ Der Begriff ist in verschiedenen Vorarbeiten eingeführt und wird von einer, vgl. Berlin Declaration on NFDI Cross Cutting Topics, <http://doi.org/10.5281/zenodo.3457213> und Leipzig-Berlin-Erklärung zu NFDI-Querschnittsthemen der Infrastrukturentwicklung, <http://doi.org/10.5281/zenodo.3895209>

¹⁰ <https://martinfowler.com/articles/data-mesh-principles.html>

Sustainable structures (Z3)

A “Common Infrastructures” section must ensure that structures are created in the NFDI in which the provision of shared information infrastructures can be organized in the long term.

A first step towards this goal was the establishment of a functioning internal organization of the section. Working groups were formed within the section to work on cross-cutting issues. The section brings together stakeholders from 26 consortia at different stages of development. Many consortia are driving forward the consolidation of existing services in their respective domains, i.e. there is work of a similar nature in the task areas of the NFDI consortia.

The members of the working groups are recruited from the NFDI consortia. If necessary, other specialist partners are also included to contribute their expertise and services. According to the “principle of best suitability”, co-optations can be made for specific tasks or services - with additional funding if necessary. A participatory culture and iterative working methods have helped to integrate players from the second and third cohorts of the NFDI consortia and to respond appropriately to parallel developments outside the NFDI.

The working groups develop analyses, concepts and concrete proposals for solutions, which are negotiated in the section as the basis for draft resolutions in the consortium meeting (see Z1). The strengths, weaknesses, opportunities and risks are regularly taken into account. In particular, workflows have been created since 2023 in close cooperation with Base4NFDI, which serve both quality assurance and sustainability.

With increasing clarification of the needs and requirements for basic services, technology partnerships for the operation of such services can be formed within the section respectively as a result of its work. To support this, two specific work results in particular are envisaged.

- A concept to provide the various architectural approaches in the NFDI with a common operational framework - based on the results of the section's “Overall Architecture” working group.
- Complementary a concept of the NFDI Task Force “Governance & Sustainability” on compliance, management and sustainability of necessary resources for common infrastructures in the NFDI, which also includes legal and contractual frameworks.

Success criteria

The objectives presented here are interlinked. They are designed for the long term and place high demands on cooperation in the NFDI, but can also build on the particular strengths of the NFDI and open up great potential for the way in which data and computing capacities can be used in science in the future. However, inherent weaknesses and risks must also be taken into account:

- Strengths
 - Despite their heterogeneity, each consortium reflects the requirements of the entire NFDI for generic services and brings with it corresponding experience, usually gained over many years.
 - Enormous capacities and a large number of services are available in the NFDI as a whole.

- A large number of standards and conceptual templates and preliminary work are available internationally (in particular EOSC).
- A lean approach is favored - the resources and services required to set up federated infrastructures are manageable.
- Weaknesses
 - Many stakeholders with a high need for coordination, resulting in a lack of agility.
 - Too few experienced service providers and a lack of personnel (especially in the multi-cloud area).
 - Asynchrony of a federated infrastructure to be set up over the entire NFDI in the longer term (investment and personnel planning, sustainability) and the current needs and developments in the consortia.
- Opportunities
 - Creation of a federated space for the efficient mobilization and use of data.
 - Visibility and networking of the NFDI as a whole, both nationally and internationally - especially as a **national EOSC node**.
- Risks
 - No long-term funding.
 - Lack of support and implementation by the consortia.
 - Overall architecture not flexible enough; no room for innovations, dynamic adaptations, domain-specific requirements or connection to international developments.
 - Lack of governance for the operation of the federated infrastructure.

Tasks

Cross-cutting tasks of the section and its working groups

(A1) - Further development Research Data Commons

One of the Section's main tasks is to promote the development of a Research Data Commons (RDC) for the NFDI. This includes identifying core components, services and standards that can be used across the NFDI, coordinating the corresponding workflows and interfaces within and between the consortia and beyond in an international context. It is important to continuously record the technical and organizational requirements for jointly usable infrastructures and the requirements for compliance, management and sustainability of the necessary technical resources. An architecture concept for the NFDI ("Overall Architecture" working group) will create the necessary framework for this work. The basic services initiated by the section can also be better positioned in terms of the strategic orientation of the NFDI. Furthermore, it is planned to use the architecture concept for content alignment and work-sharing planning within and between the WGs as well as with the other sections and the NFDI TF "Governance and Sustainability". The consortia will be involved via presentations and reports at consortium and spokesperson meetings, via the section spokespersons, the WG coordinators and the members of the section and the WGs.

(A2) - Governance of the Section and Coordination of the working groups

- Establishment, evaluation and dissolution (status *inactive*) of WGs
- Preparation and implementation of section meetings
- Presentation of the results from the WGs at section meetings
 - Development in WGs and presentation at section meetings
 - Discussion of work progress
- Presentation of the basic services of the WGs at section meetings
 - Supervision of basic services in WGs and presentation at section meetings
 - Discussion of proposals
 - Discussion of work progress
- Regular exchange with WGs, other sections and consortia
- Regular reporting to the Directorate and Consortium Assembly
- Contribution to evaluations and follow-up proposals of the consortia, as well as to structural evaluation
- Process optimization, if necessary also by acquiring financial and personnel support

(A3) - Development of internal documentation and knowledge base

- Development of an internal knowledge representation
- Overview of contact persons in the working groups and consortia for specific topics
- Overview of internal activities, in particular upcoming dates and topics currently being worked on
- Documentation of the results of the section and the working groups

(A4) - Development of an outreach strategy and coordination of outreach activities

- Development of an external knowledge representation
- Overview of external activities, in particular upcoming dates and topics currently being worked on
- Publication of the results of the section and the working groups
- International networking through invited lectures and joint events
- Contribution to CoRDI (NFDI Conference) and B4UC (Base4NFDI User Conference)
- Contribution to political debates, such as position papers on the Research Data Act and the Data Institute

Tasks of the individual working groups

The working groups pursue individual goals, which are briefly summarized below. For further details, please refer to the (continuously updated) concepts of the individual working groups. It should be pointed out here that sustainability must be kept in mind for all activities of the working groups and the basic services that are created.

Overall Architecture (OA)

A continuously and jointly developed overall architecture creates a binding framework with the necessary “guard rails” for further technical and organizational developments within the NFDI and enables targeted and practical accents for improved national and international networking and communication. The development of an overall architecture for the NFDI is a complex task. It requires cooperation, flexibility and adaptability in order to meet the different needs of the participating consortia. The individual steps should be clearly defined and manageable. Due to the dynamic nature of IT-supported infrastructures, an iterative approach that keeps an eye on all possible steps is recommended.

Contact: Marius Politze, Philipp Wieder

Data Integration (DI)

The immense heterogeneity of scientific data and the associated challenges in data integration often hinder the derivation of new insights, particularly in the context of interdisciplinary projects. The goal of this working group is therefore to develop and provide scalable methods for integrating data and metadata to enable easy data exchange between users and services. In particular, physical and virtual integration methods, methods for schema matching and semantic annotation, as well as knowledge graphs are considered. Additionally, based on concrete user stories, an architecture for the integration of scientific data is developed and a first reference implementation is provided.

Contact: Bernhard Seeger; Andreas Henrich; Thorsten Papenbrock; Dirk Riehle

Data Management Planning (DMP)

The working group aims to coordinate the DMP activities of the NFDI consortia. This includes analogous concepts such as the Software Management Plan (SMP). In addition to establishing DMP tools, coordinating their further development, and harmonizing the content of DMP templates, concepts for the strategic implementation of DMPs in research workflows, as well as training and consulting services, are being developed. Since DMPs are essential connecting or intermediate pieces of the data management workflow and cover almost the entire breadth of its content, the working group aims for partial collaboration with the other NFDI sections, such as the Edutrain and Metadata sections..

Contact: Daniela Hausen; Jürgen Windeck

Data Science and Artificial Intelligence (DSAI)

The various NFDI consortia are developing tools and services based on innovative methods such as the use of large language models (LLMs). The working group serves as a forum for exchanging information about these tools and services and aims to increase their visibility both within and beyond the NFDI. By establishing these tools and services, the working group

aims to enhance the adoption of the infrastructures provided by the NFDI consortia within the scientific community.

Contact: Adamantios Koumpis; Arnim Bleier; Sonja Schimmler

Electronic Lab Notebooks (ELN)

The working group comprises representatives from various NFDI consortia, with a focus on the natural sciences and life sciences. Its aim is to prepare and establish services that support the selection of suitable electronic laboratory notebooks (ELNs) and their implementation within consortia, their communities, and individual groups. The working group will systematically evaluate various ELNs and their data structures. Furthermore, a platform will be established for testing different ELNs and similar (embedded) tools, to ensure continued access to relevant solutions and enable comparative analysis.

Contact: Nicole Jung; Christof Wöll; Felix Bach

Identity and Access Management (IAM)

The working group aims to develop processes, policies, and architectures to simplify the management of digital identities and federated access to resources within and between NFDI consortia. To this end, existing and emerging IAM systems need to be connected and extended in a way that enables researchers from various fields and institutions to access NFDI resources easily and securely. Interoperability is a central aspect here, both within the NFDI and with regard to access to and exchange with external infrastructures such as the European Open Science Cloud or the Life Science AAI.

Contact: Marius Politze; Wolfgang Pempe

Infrastructure and Data Security (IDS)

Scientific disciplines often involve the processing of personal data subject to the EU General Data Protection Regulation (GDPR) or other sensitive data types. These require technical and organizational measures to ensure a high level of data security. The working group focuses on the necessary technical and organizational safeguards that must be implemented in the research infrastructure to ensure the secure and lawful processing and management of personal and sensitive data.

Contact: Petra Ritter

Long-term Archival (LTA)

The working group aims to identify and harmonize methods, procedures, and techniques for long-term preservation and use of data and software as reproducible information objects. The data should be preserved at least as a bitstream, but ideally also be made accessible across software, hardware, and technological changes. The working group comprises members from various NFDI consortia in the humanities, natural sciences, and engineering, as well as experts from various relevant institutions and infrastructures. It is closely linked to nestor¹¹, the

¹¹ https://www.langzeitarchivierung.de/Webs/nestor/EN/Home/home_node.html

competence network for long-term archiving. This close cooperation ensures that both the current state of the art in long-term archiving is represented in the group and that the knowledge of data producers about the creation contexts and the needs of data users is directly integrated.

Contact: Peter Leinen, Katharina Markus, Thomas Stäcker

Multi-Cloud (MC)

A key objective of all NFDI consortia is the systematic registration and sustainable provision of data to enable open and uniform access at a national and even international level. The foundation for this is the concept of a federated multi-cloud architecture. This architecture provides unified access to computing and data storage resources through a federated identity and access management (IAM) infrastructure and enables seamless integration of distributed, heterogeneous services and data at a higher level.

Contact: Alexander Sczyrba, Kilian Schwarz

Persistent Identifiers (PID)

Persistent Identifiers (PIDs) are an essential component of FAIR data management. The working group serves as a link between the PID4NFDI basic service established in 2024, the section, and the individual consortia. It brings together PID service providers and users and supports them in identifying the specific needs of the respective communities for an NFDI-wide PID service and in evaluating the developed service components accordingly. In addition, it serves as a forum for exchange with other working groups within the section and for identifying cooperation opportunities on cross-working group and cross-service topics such as information and training on the use and integration of PIDs, metadata quality assurance, and the development of interfaces between PID services and user systems.

Contact: Sven Bingert, Stephanie Hagemann-Wilholt

Research Software Engineering (RSE)

The working group networks the NFDI consortia in software-related aspects. It focuses on three main areas: research software, software communities, and NFDI software infrastructure. In an advisory and supportive role, the working group operates a central forum and establishes the required software ecosystem within NFDI for the professional development of software infrastructure components, which together form an integral part of the NFDI. In addition, the working group acts as an interface between the NFDI and comparable European and international initiatives to promote the connectivity of the NFDI with other infrastructures.

Contact: Florian Thiery; Bernd Flemisch

Timeline with Milestones

The section is designed for long-term existence. The work program was initially outlined for 24 months and is continuously being developed. The following is the plan for the next 24 months.

An iterative process is also being pursued, meaning that results are regularly presented and discussed.

Milestone plan

<i>Milestone</i>	<i>Time</i>	<i>Description</i>
M.1	Jul. 2025	Consolidation of working groups and continuation of individual work programs
M.2	Jul. 2025	Draft concept of overall architecture
M.3	Jul. 2026	Publication of the concept of overall architecture
M.4	Jul. 2025	Initiation of an internal knowledge base (iterative continuation)
M.5	Dec. 2025	Initiation of an external knowledge representation (iterative continuation)
M.6	Dec. 2026	Federated NFDI basic infrastructure as a "Minimum Viable Product"

Cooperation within NFDI

A key instrument for cooperation within the NFDI are the working groups of the section. The members of the working groups ensure the exchange with the **consortia** and promote coordination between the work in the consortia and the section's working groups.

Since the beginning of the setup phase, there has also been a lively, initially predominantly bilateral exchange between the **sections**. Since 2023, there have also been regular meetings of the section spokespersons with the directorate. This format of dialogue contributes significantly to the developments in the sections, but also to the overall strategic direction.

There are close interconnections with the section "Metadata, Terminologies, Provenance". In line with the objectives of the "Common Infrastructures" section, this section focuses more on semantic interoperability and the appropriate structures for this. Coordination also takes place at the working group level here.

Coordination or exchange of assessments with the section "Ethical, Legal and Social Aspects" is also important. Compliance or legal questions regarding the operation of a collaborative multi-cloud-based environment cannot be fully resolved by the predominantly technical staffing of the section, even if several partners bring relevant experience and expertise from the operation of their services or other projects.

Training and education offerings are an important part of a federated infrastructure. This applies to both materials relating to the use of the infrastructure and materials for the reuse of data. Of particular interest is the training platform Dalia¹², which is being promoted by the "Training & Education" section.

¹² <https://dalia.education/>

With the recently established section "Industry Engagement", cooperative fields are also opening up, especially in the area of cloud-based architectures. The requirements for infrastructure developments in commercial companies and academia show many convergences. Specifically, the "Common Infrastructures" section is indirectly involved in the BMBF-funded GAIA-X project FAIR Data Spaces via some consortia.

The section and its working groups make a significant contribution to Base4NFDI. Already in the application phase, the section made an integral contribution, and thus also shaped the role definitions and workflows. Basic services are prepared in the section's working groups, discussed by the section and its members and, if positive, approved. An important point in the discussion is always the comparison with the targeted overall architecture of the NFDI. The development of basic services is accompanied by the working groups in all phases (initialization, integration, ramping up for operation) and is evaluated and re-coordinated by the section from phase to phase. It should be emphasized that the setup of basic services, in particular for the integration and ramping up for operation phase, must be provided by the consortia for the most part. The section's working groups and their members form an important, mediating hinge.

Cooperation with initiatives outside NFDI

The section is involved in various initiatives outside of NFDI. Many actors not only contribute to the NFDI but are also active in other (inter)national alliances (e.g., Open Source and Linked Data Community), standardization initiatives and projects (e.g., World Wide Web Consortium). It is desirable to both apply standards and specifications from these initiatives and proactively contribute NFDI developments to the respective communities. Some highlights illustrate this:

Central initiatives

In the EOSC - European Open Science Cloud, structures have already been formed that address comparable questions of harmonizing data infrastructures - this includes both very concrete implementation projects (especially EOSC Future¹³), as well as various task forces that negotiate agreements on relevant fundamental issues. The NFDI has links to this central European infrastructure project on several levels: as an overall organization, through the subject-specific consortia and through the sections. In the section, attention must be paid to incorporating relevant results from the technical standardization groups and avoiding duplication or competing requirements for the NFDI consortia. In addition, the section can contribute to an overall strategy for connecting NFDI and EOSC. The section supports the establishment of the NFDI as a national EOSC node¹⁴; in particular by providing an EOSC platform with federated basic services.

At the European level, there is also important preparatory work in the ESFRI Landmarks or projects; also with regard to the application of agreed standards or the use of common basic services. In the section, some of these and other domain-specific specifications are discussed,

¹³ <https://eoscfuture.eu/>

¹⁴ <https://eosc.eu/news/2023/09/geant-publishes-eosc-nodes-position-paper-on-behalf-of-nren-community/>

in some cases there will be an interaction to make services compatible on both sides. The same applies to the global Research Data Alliance¹⁵.

Several NFDI consortia are already planning to use a "Digital Objects" architecture, which is currently being specified in a global initiative, the FAIR Digital Objects Forum^{16 17}. This concept must be discussed in the context of the work on Common Infrastructures and integration possibilities must be found.

For the development of cloud-based services such as the Research Data Commons, there are significant synergies through the FAIR Data Spaces project¹⁸ with GAIA-X and with Data Spaces in the general sense. With the substantial participation of some NFDI consortia, both technical work has been carried out here, which can be reused in the section, as well as fundamental legal clarifications and the use of common standards with industry have been addressed.

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¹⁵ <https://rd-alliance.org>

¹⁶ <https://fairdo.org>

¹⁷ Smedt, Koenraad de; Koureas, Dimitris; Wittenburg, Peter (2020): FAIR Digital Objects for Science: From Data Pieces to Actionable Knowledge Units. In: Publications 8 (2), S. 21. DOI: 10.3390/publications8020021.

¹⁸ <https://www.nfdi.de/fair-data-spaces/>

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Participating members in the section

In the section, a multitude of people from the consortia and beyond are involved. An overview can be found on the website: nfdi.de/section-infra